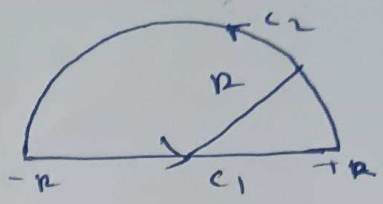
CT-2

① $$I = \frac{1}{2}\Big[\underbrace{\int_{-\infty}^{\infty} \frac{\cos ax}{1+x^2}\,dx}_{I_1} + \underbrace{\int_{-\infty}^{\infty} \frac{x \sin ax}{1+x^2}\,dx}_{I_2}\Big].$$

$$\therefore I_1 = \mathrm{Re}\Big[\lim_{R\to\infty} \int_{-R}^{R} \frac{e^{iax}}{1+x^2}\,dx\Big]$$

$$= \mathrm{Re}\Big[\oint_C \frac{e^{iaz}}{1+z^2}\,dz - \int_{C_2} \underbrace{\frac{e^{iaz}}{1+z^2}}_{f(z)}\,dz$$



$C = C_1 + C_2$

$a > 0$:

we close C_2 in the upper-half of $\mathbb{C}$.

$$\int_{C_2} f(z)\,dz = 0 \quad [\text{Jordan's lemma}]$$

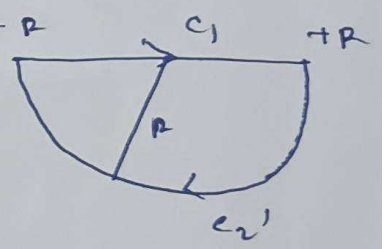
$f(z)$ has a pole @ $z_0 = i$ as we are in upper $\mathbb{C}$.

$$\oint_C f(z)\,dz = 2\pi i \;\mathrm{Res}\, f(z)\big|_{z=i}$$

$$= 2\pi i \;\frac{e^{iaz}}{z+i}\Big|_{z=i} = \boxed{\pi e^{-a}} \quad a > 0$$

$a < 0$:

we close C_2' in the lower-half of $\mathbb{C}$.



$$\int_{C_2'} f(z)\,dz = 0 \quad [\text{Jordan's lemma}]$$

$C' = C_1 + C_2'$

$f(z)$ has a pole @ $z_0 = -i$ as we are in lower $\mathbb{C}$.

$$\therefore \oint_{C'} f(z)\,dz = -2\pi i \;\mathrm{Res}\, f(z)\big|_{z=-i} \quad [\text{clockwise direct}^n]$$

$$= -2\pi i \;\frac{e^{iaz}}{z-i}\Big|_{z=-i} = \boxed{\pi e^{a}}$$

$$\therefore \left.\begin{array}{ll} a > 0 & \pi e^{-a} \\ a < 0 & \pi e^{a} \end{array}\right] \longrightarrow I_1.$$

(only real part comes as the answer so no need to take Re)

$$\therefore \boxed{I_1 = \pi e^{-|a|}}$$

$\therefore I_2 = \int_{-\infty}^{\infty} \frac{x \sin(ax)}{1+x^2}\, dx.$

Now; $\frac{dI_1(a)}{da} = \frac{d}{da}\int_{-\infty}^{\infty} \frac{\cos ax}{1+x^2}\, dx = -\int_{-\infty}^{\infty} \frac{x \sin ax}{(1+x^2)}\, dx$

$= -I_2.$

$\therefore I_2 = -\frac{dI_1(a)}{da} = -\frac{d}{da}\left[\pi e^{-|a|}\right]$

$\underline{a>0}$: $I_2 = -\frac{d}{da}\left[\pi e^{-a}\right] = \pi e^{-a}.$

$\underline{a<0}$: $I_2 = -\frac{d}{da}\left[\pi e^{a}\right] = -\pi e^{a}$

$\therefore$ ~~$I_2(a) = \pi e^{-|a|}$~~

$I = \frac{1}{2}\left[I_1 + I_2\right]$

$\underline{a>0}$: $I_1 = \pi e^{-a}$; $I_2 = \pi e^{-a}$

$\boxed{I = \pi e^{-a}}.$

$\underline{a<0}$ $I_1 = \pi e^{a}$; $I_2 = -\pi e^{a}$

$\boxed{I = 0}.$

$\therefore I = \begin{cases} \pi e^{-a} & ; a>0. \\ 0 & ; a<0 \end{cases}$ [Ans]

$\underline{a=0}$:

$I = \int_0^{\infty} \frac{dx}{1+x^2} = \int_0^{\infty} \frac{dx}{1+x^2} = \tan^{-1}(x)\Big|_0^{\infty} = \frac{\pi}{2}$ (Ans)

③ Done Previously in HW-6.

② $f(z) = \frac{1}{z(\sin(\pi z))(z+\frac{1}{2})}.$

~~Locate~~ Poles occur @ $z=0$; $z=-\frac{1}{2}$ &

$\sin \pi z = 0$ ie $z = n$ $[n \in \mathbb{Z}]$

@ $z \to 0$; $\sin \pi z \approx \pi z$

$\therefore f(z) = \dfrac{1}{z(\pi z)(\frac{1}{2})} = \dfrac{2}{\pi z^2}$

$\therefore z = 0 \longrightarrow$ Pole of order 2.

@ $z \to -\frac{1}{2}$; only $(z+\frac{1}{2})$ becomes 0.

$\longrightarrow$ Pole of order 1.

@ $z \to n \neq 0$; only $\sin \pi z$ vanishes. $\longrightarrow$ Pole of order 1.

@ $\underline{z = 0}$: (o.n = 2)

$$\text{Res } f(z)\Big|_{z=0} = \lim_{z\to 0} \frac{d}{dz}\left(z^2 f(z)\right).$$

$$= \lim_{z\to 0} \frac{d}{dz}\left[\frac{z}{(z+\frac{1}{2})\sin \pi z}\right]$$

$$= \lim_{z\to 0} \frac{d}{dz}\left[\frac{\cancel{z}}{(z+\frac{1}{2})(\pi \cancel{z})}\right]$$

$$= \lim_{z\to 0} \cancel{d} - \frac{1}{\pi}\cdot\frac{1}{(z+\frac{1}{2})^2}\Big|_{z=0}.$$

$$\boxed{\text{Res } f(z)\Big|_{z=0} = -\frac{4}{\pi}}$$ (Ans)

@ $\underline{z = -\frac{1}{2}}$ (n=1)

$$\text{Res } f(z)\Big|_{z=-\frac{1}{2}} = \lim_{z\to -\frac{1}{2}} (z+\tfrac{1}{2})\, f(z)$$

$$= \lim_{z\to -\frac{1}{2}} \frac{1}{\sin(\pi z)\, z} = \frac{-2}{-\sin\frac{\pi}{2}} = \boxed{2 = \text{Res } f(z)\Big|_{z=-\frac{1}{2}}}$$

(Ans)

@ $\underline{z = n \neq 0}$ (n = 1 $\rightarrow$ order)

$$\text{Res } f(z)\Big|_{z=n} = \lim_{z\to n} (z-n)\, f(z)$$

$$= \lim_{z\to n} (z-n)\frac{1}{z(z+\frac{1}{2})\sin \pi z}$$

Let $z = n + \epsilon$; $\epsilon \to 0$

$\therefore \sin(\pi z) = \sin \pi(n+\epsilon) = \sin n\pi \cos \pi\epsilon + \cos n\pi \sin \pi\epsilon$

$= (-1)^n \sin \pi\epsilon$ [$\sin n\pi = 0$. $\cos n\pi = (-1)^n$]

$\approx (-1)^n \pi\epsilon$.

$$\boxed{\sin(\pi z) = \pi(-1)^n (z-n)}$$

$$\therefore \operatorname{Res} f(z)\Big|_{z=n} = \lim_{z \to n} \cancel{(z-n)} \cdot \frac{1}{z\left(z+\frac{1}{2}\right)\pi(-1)^n \cancel{(z-n)}}$$

$$= \frac{1}{n\left(n+\frac{1}{2}\right)\pi(-1)^n}$$

$$\boxed{\operatorname{Res} f(z)\Big|_{z=n} = \frac{(-1)^n}{\left(n+\frac{1}{2}\right) n \pi}}$$ (Ans)